

Real-Branch Asymptotic Reversion

with Logarithms and Irrational Powers

A constructive coefficient calculus, rigorous remainder bounds,
and the Wolfram Language package `PowerLogInverse`

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For

$$f(x) = ax^p \left(1 + \sum_{i=1}^s x^{\delta_i} P_i(\log x) \right), \quad a, p, \delta_i > 0,$$

put $u = (y/a)^{1/p}$ and $L = \log u$. The positive inverse tending to zero is

$$f^{-1}(y) = u \left(1 + \sum_{\mathbf{m} \neq 0} u^{\mathbf{m} \cdot \boldsymbol{\delta}} Q_{\mathbf{m}}(L) \right),$$

where, with $n = |\mathbf{m}|$ and $\Delta = \mathbf{m} \cdot \boldsymbol{\delta}$,

$$Q_{\mathbf{m}}(L) = \frac{(-1)^n}{p^n \prod_i m_i!} \prod_{j=1}^{n-1} \left(\frac{d}{dL} + 1 + \Delta + pj \right) \prod_i P_i(L)^{m_i}.$$

Every finite power cutoff is effectively computable. The logarithmic factors in its remainder are tracked rather than hidden in an ordinary power-series order symbol.

Abstract

The inverse of $x + x^2(1 + \log x)$ on the positive real axis has logarithmic coefficient polynomials, whereas the inverse of $x + x^{\sqrt{2}}$ has an irrational exponent progression. Neither example should be treated as an unspecified analytic inverse at an arbitrary zero of a complexified function. We explain the complex Taylor germ recorded in the motivating Mathematica question, then construct the desired real germs directly. A specialization of the classical Lagrange–Bürmann formula gives explicit multi-index differential polynomials for finite sums of power–logarithmic corrections to ax^P . A finitely generated positive exponent semigroup makes strict cutoff computation finite even when the corresponding additive group is dense. An analytic implicit-function argument proves actual convergence for exact finite input of this class, and also gives a computable asymptotic remainder bound for every cutoff. Residuals transfer to inverse errors through an explicit derivative estimate. The two original examples are developed to arbitrary order; the irrational-power example also has a closed coefficient formula and an exact convergence radius. Extensions include observable inversion, leading logarithmic pivots with real Lambert- W branches, uncertain input jets, and flat perturbations. The accompanying package implements the finite power–logarithmic core, exact algebraic support ordering, branch-aware normalization, and an independent truncated residual calculation. Mathematical validation and native-kernel verification are reported separately.

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1 The question is about a germ, not merely an inverse symbol

The archived question [1] asks for the inverse near $y = 0^+$ of

$$f_1(x) = x + x^2(1 + \log x), \quad x > 0, \quad (1.1)$$

and, in its update, for the corresponding inverse of

$$f_2(x) = x + x^{\sqrt{2}}, \quad x > 0. \quad (1.2)$$

We use x for the source variable and y for the argument of the inverse. This avoids confusing a series for f with a series for f^{-1} . All logarithms of positive quantities below are real logarithms.

The request has three logically distinct components: identify the intended inverse branch, identify a scale closed under the necessary operations, and compute coefficients with a meaningful remainder. Merely attaching $y > 0$ to an unspecified inverse expression addresses none of these components completely. An inverse at a complex zero can be evaluated for small positive y and still remain complex.

1.1 The real branches exist globally in both examples

For the first function,

$$f_1'(x) = 1 + x(3 + 2 \log x).$$

The minimum of $x(3 + 2 \log x)$ occurs at $x = e^{-5/2}$ and equals $-2e^{-5/2}$. Consequently,

$$f_1'(x) \geq 1 - 2e^{-5/2} > 0 \quad (x > 0). \quad (1.3)$$

Together with $f_1(x) \rightarrow 0$ as $x \rightarrow 0^+$ and $f_1(x) \rightarrow \infty$ as $x \rightarrow \infty$, this proves that f_1 is a bijection of $(0, \infty)$ onto itself. For f_2 , the same conclusion follows from $f_2'(x) = 1 + \sqrt{2}x^{\sqrt{2}-1} > 0$ and the endpoint limits. Thus the real inverse is unambiguous in these examples. The problem is not a lack of a real branch; it is how to represent and select it.

1.2 Why the archived output is a correct complex Taylor series

The archived computation returned a constant involving

$$w = W_{-1}(-e), \quad z_0 = e^{w-1}.$$

Since $-e < -1/e$, this value of W_{-1} is not real. On the compatible local logarithm branch, $\log z_0 = w - 1$. Also $we^w = -e$ gives $z_0w = -1$, hence

$$f_1(z_0) = z_0(1 + z_0(1 + \log z_0)) = 0.$$

The derivative identities simplify to

$$f_1'(z_0) = z_0 - 1, \quad f_1''(z_0) = 3 + 2w.$$

The ordinary analytic inverse at this nonzero complex zero therefore has

$$G(y) = z_0 + \frac{y}{z_0 - 1} - \frac{3 + 2w}{2(z_0 - 1)^3}y^2 + O(y^3), \quad (1.4)$$

exactly explaining the structure of the recorded output. This is a valid local inverse calculation at the wrong germ for the requested application. Other Lambert- W indices require attention to the logarithm sheet; one must not indiscriminately treat every such index as a zero for a fixed principal logarithm.

Our construction selects a different germ from the outset:

$$g(y) > 0, \quad g(y) \longrightarrow 0, \quad \frac{g(y)}{y} \longrightarrow 1. \quad (1.5)$$

For the general leading term ax^p , the last condition becomes $g(y)/(y/a)^{1/p} \rightarrow 1$. This asymptotic address is part of the mathematical problem, not a formatting assumption appended to an already chosen branch.

The archive is evidence about a particular recorded computation, not a benchmark of every current Wolfram Language version. The package avoids reliance on the branch-selection behavior of `InverseFunction`. The official documentation describes that symbol and its treatment of inverse branches [7]; no new claim about a current built-in failure is needed here.

1.3 The answers in their natural scales

Writing $L = \log y$, the first inverse begins

$$g_1(y) = y - (L+1)y^2 + (2L^2 + 5L + 3)y^3 - \left(5L^3 + \frac{41}{2}L^2 + 27L + \frac{23}{2}\right)y^4 + O(y^5(1+|L|)^4). \quad (1.6)$$

In particular, the tempting two-term notation $y - (1 + \log y)y^2 + O(y^3)$ is *not* a valid remainder for this real inverse. The correct two-term estimate is

$$g_1(y) = y - (1 + \log y)y^2 + O(y^3 \log^2 y). \quad (1.7)$$

This observation concerns the inverse remainder; an order symbol used to truncate the *input* function is a separate matter.

For $r = \sqrt{2}$, the second inverse begins

$$\begin{aligned} g_2(y) &= y - y^r + ry^{2r-1} - \frac{r(3r-1)}{2}y^{3r-2} + O(y^{4r-3}) \\ &= y - y^{\sqrt{2}} + \sqrt{2}y^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2}y^{3\sqrt{2}-2} + O(y^{4\sqrt{2}-3}), \end{aligned} \quad (1.8)$$

which is the four-block expansion requested in the update.

The first inverse has a C^1 extension to zero but not a C^2 one: $g_1''(y) \sim -2 \log y \rightarrow +\infty$. For the second, $g_2''(y) \sim -r(r-1)y^{r-2} \rightarrow -\infty$. Ordinary Taylor expansions at zero cannot encode these behaviors. Ordinary Puiseux series on a fixed rational power grid do not encode the irrational progression either.

2 Power-log blocks and cutoff-finite support

2.1 The representation

A power-log block is an expression

$$u^\gamma Q(L), \quad L = \log u, \quad Q \in \mathbb{R}[L].$$

We keep a whole logarithmic polynomial at a given power together. For fixed k, ℓ and $\gamma < \eta$,

$$u^\eta (\log u)^\ell = o(u^\gamma (\log u)^k) \quad (u \rightarrow 0^+),$$

provided the denominator monomial is nonzero. Thus the exponent is the primary order coordinate; within a block the largest nonzero logarithmic degree determines its leading magnitude. Odd logarithmic powers can be negative, so magnitude estimates use $|\log u|$.

Let $\delta_1, \dots, \delta_s$ be fixed positive real numbers and define

$$\Gamma = \left\{ \sum_{i=1}^s m_i \delta_i : m_i \in \mathbb{N} \right\}. \quad (2.1)$$

The formal relative corrections live in locally finite sums $\sum_{\gamma \in \Gamma} u^\gamma Q_\gamma(L)$. For the inverse itself the support lies in $1 + \Gamma$.

Lemma 2.1 (Local finiteness). *For every finite B , only finitely many multi-indices satisfy $\mathbf{m} \cdot \boldsymbol{\delta} < B$. In particular, $\Gamma \cap [0, B)$ is finite.*

Proof. Every such index satisfies $0 \leq m_i < B/\delta_i$. It is therefore contained in a finite integer box. Different indices may have the same weight, which only reduces the number of distinct support points. $\square$

This distinction is important. The additive *group* generated by two incommensurable real numbers can be dense, but our *positive semigroup* is locally finite. Irrationality does not create infinitely many terms below a fixed cutoff. It can create successively smaller gaps far out in the support; no global lower bound on those gaps is assumed.

2.2 Formal arithmetic and composition

For a nonzero formal sum, define its power valuation to be its least exponent. Multiplication adds valuations. A logarithmic polynomial is a coefficient and has power valuation zero. Differentiation obeys

$$\frac{d}{du}(u^\gamma Q(\log u)) = u^{\gamma-1}(\gamma + D_L)Q(L), \quad D_L = \frac{d}{dL}. \quad (2.2)$$

One may enlarge the ambient exponent group when differentiating, but inside every algorithm below only a controlled translate of Γ is used.

If V has positive valuation, the formal expansions

$$\begin{aligned} (1 + V)^q &= \sum_{k \geq 0} \binom{q}{k} V^k, \\ \log(1 + V) &= \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} V^k \end{aligned} \quad (2.3)$$

are cutoff-finite: only finitely many powers V^k can affect a prescribed cutoff. The substitution $x = u(1 + V)$ is consequently well defined by

$$x^\delta = u^\delta (1 + V)^\delta, \quad \log x = L + \log(1 + V).$$

These are identities on the positive real branch when the formal sums are realized by sufficiently small actual corrections.

2.3 Why not manufacture a SeriesData object?

The documented SeriesData representation indexes powers by n/den [6]. A custom sparse list of exact exponents and logarithmic coefficient polynomials makes the intended support and cutoff explicit without relying on that representation to perform nonstandard reversion. The package therefore returns ordinary finite expressions together with structured metadata. Its remainder object is an inert descriptor, not a counterfeit SeriesData object.

This finite-support viewpoint is compatible with broader transseries frameworks [5]. The supplied ProveIt draft corpus [2] also emphasizes support-aware operations, real-germ selection, and the separation between formal calculation and analytic error transfer. Those principles are useful here without importing an entire general transseries field into the implementation.

3 Prepared finite germs and the selected real inverse

Definition 3.1 (Prepared finite power-log germ). A prepared finite germ is

$$f(x) = ax^p(1 + H(x)), \quad H(x) = \sum_{i=1}^s x^{\delta_i} P_i(\log x), \quad (3.1)$$

where $a, p > 0$, every $\delta_i > 0$, and every P_i is a real polynomial. Equal exponents may be combined and zero polynomials removed. The case $s = 0$ is allowed and is an exact monomial.

The theory permits arbitrary positive real exponents. The executable package restricts support exponents to exact real algebraic numbers so that order and equality decisions are exact and operationally meaningful.

Since a positive power beats every fixed logarithmic power,

$$H(x) \rightarrow 0, \quad xH'(x) \rightarrow 0.$$

Consequently,

$$f'(x) = ax^{p-1}(p(1 + H(x)) + xH'(x)) \sim apx^{p-1}. \quad (3.2)$$

Proposition 3.2 (Existence and branch address). *Every prepared finite germ has a unique positive real inverse on some interval $(0, y_0)$ with $g(y) \rightarrow 0$. With*

$$u = \left(\frac{y}{a}\right)^{1/p} > 0, \quad L = \log u = \frac{1}{p} \log \frac{y}{a}, \quad (3.3)$$

one has $g(y)/u \rightarrow 1$.

Proof. Equation (3.2) makes f strictly increasing and positive on a sufficiently short interval $(0, x_0)$. Its limit at zero is zero, so it maps this interval bijectively onto $(0, f(x_0))$. If $x = g(y)$, then $u = x(1 + H(x))^{1/p}$, whence $x/u \rightarrow 1$. $\square$

This is an eventual local assertion. Unlike the two motivating examples, a general prepared finite germ need not be globally monotone or injective. For example, a large negative correction can create turning points away from zero. The package returns the addressed small positive branch and does not claim to classify remote branches.

Remark 3.3 (Cutoff convention). A package cutoff $C > 1$ means: retain inverse blocks whose *power of u* is strictly less than C . In the original target variable this is a strict y -power cutoff C/p . Thus for $p \neq 1$ a cutoff number must not be read as an exponent of y . The normalization rules are included in the returned data specifically to expose this distinction.

4 An explicit multi-index reversion formula

For $\mathbf{m} = (m_1, \dots, m_s) \in \mathbb{N}^s$, put

$$n = |\mathbf{m}| = \sum_i m_i, \quad \Delta = \sum_i m_i \delta_i, \quad \mathbf{m}! = \prod_i m_i!, \quad P^{\mathbf{m}}(L) = \prod_i P_i(L)^{m_i}.$$

An empty product of differential operators is the identity.

Theorem 4.1 (Power–logarithmic inverse coefficients). *The addressed inverse of (3.1) has the expansion*

$$g(y) = u \left(1 + \sum_{\mathbf{m} \in \mathbb{N}^s \setminus \{0\}} u^\Delta Q_{\mathbf{m}}(L) \right), \quad (4.1)$$

where

$$Q_{\mathbf{m}}(L) = \frac{(-1)^n}{p^n \mathbf{m}!} \left[\prod_{j=1}^{n-1} (D_L + 1 + \Delta + p j) \right] P^{\mathbf{m}}(L). \quad (4.2)$$

For exact finite input this series converges absolutely for all sufficiently small positive u . As a formal series it is the unique inverse with leading term u . Terms from different multi-indices with the same Δ must be added before an actual nonzero support is reported.

The coefficient identity is a specialization of classical Lagrange–Bürmann inversion, not a claim of a new general inversion theorem. Its useful features here are the explicit logarithmic differential operators, the real normalization, and a cutoff mechanism adapted to irrational support.

4.1 Homotopy form of Lagrange–Bürmann

Let $Z + \tau A(Z) = Y$, and let Φ be analytic near a nonzero base value Y . The inverse germ satisfying $Z = Y$ at $\tau = 0$ obeys

$$\Phi(Z) = \Phi(Y) + \sum_{n \geq 1} \frac{(-\tau)^n}{n!} \frac{d^{n-1}}{dY^{n-1}} (\Phi'(Y) A(Y)^n). \quad (4.3)$$

This is the usual observable form of the Lagrange inversion theorem; standard formulations and residue versions are given in DLMF §1.10(vii) [3]. Importantly, the expansion variable in (4.3) is initially the homotopy parameter τ , not Y at a singular origin. For fixed positive Y , real powers and logarithms are analytic on a small complex neighborhood with the chosen branches. The uniform justification for setting $\tau = 1$ as $Y \rightarrow 0^+$ will be proved in Section 5.

4.2 Reduction to a near-identity map

Set

$$Z = x^p, \quad Y = y/a = u^p, \quad \Phi(Z) = Z^{1/p}.$$

Then $y = f(x)$ is equivalent to

$$Y = Z + A(Z), \quad A(Z) = Z \sum_i Z^{\delta_i/p} P_i \left(\frac{\log Z}{p} \right).$$

Expanding A^n by the multinomial theorem gives, for an index of total size n ,

$$\Phi'(Y) \frac{n!}{\mathbf{m}!} Y^{n+\Delta/p} P^{\mathbf{m}}(L) = \frac{n!}{p \mathbf{m}!} Y^{n-1+(1+\Delta)/p} P^{\mathbf{m}}(L).$$

The elementary derivative identity

$$\frac{d^k}{dY^k} \left(Y^b Q \left(\frac{\log Y}{p} \right) \right) = Y^{b-k} \prod_{h=0}^{k-1} \left(b - h + \frac{1}{p} D_L \right) Q(L) \quad (4.4)$$

follows by induction. Apply it with $k = n - 1$ and $b = n - 1 + (1 + \Delta)/p$. Reversing the order of the commuting factors and using $Y^{(1+\Delta)/p} = u^{1+\Delta}$ yields exactly (4.2).

The operators commute because they are all D_L plus scalar constants. They preserve polynomiality and do not increase logarithmic degree. Thus

$$\deg Q_m \leq \sum_i m_i \deg P_i. \quad (4.5)$$

Cancellation can lower the degree or annihilate a block completely.

4.3 The first two correction levels

For $m = e_i$,

$$Q_{e_i} = -\frac{P_i}{p}.$$

For a repeated correction,

$$Q_{2e_i} = \frac{(D_L + 1 + 2\delta_i + p)P_i^2}{2p^2}.$$

For $i \neq j$,

$$Q_{e_i+e_j} = \frac{(D_L + 1 + \delta_i + \delta_j + p)(P_i P_j)}{p^2}.$$

The factor $1/2$ appears only in the repeated case. It comes from $m!$, not from a blanket $n!$ after multinomial expansion. This is an easy place for an implementation to introduce a mixed-term bug.

4.4 Formal uniqueness and a triangular alternative

Write $g = u(1 + V)$ and substitute into $f(g) = au^p$. The relative equation is

$$\mathcal{F}(V) = (1 + V)^p \left[1 + \sum_i u^{\delta_i} (1 + V)^{\delta_i} P_i(L + \log(1 + V)) \right] - 1 = 0. \quad (4.6)$$

Its linear term at power valuation zero is pV . All additional terms involving V either contain a positive support weight or are at least quadratic in V . If two solutions first differ at weight γ , the coefficient of their difference in (4.6) is p times that first block. Since $p \neq 0$, the block must vanish, a contradiction.

This also gives a recursive algorithm: visit support weights in increasing order; at weight γ , evaluate the already determined residual and set the new coefficient equal to minus that residual coefficient divided by p . The closed formula and this triangular procedure solve the same formal problem by different computational routes.

5 Convergence and explicit asymptotic remainders

A formal identity does not by itself supply a real asymptotic theorem. Here the finite nature of the input makes an analytic proof particularly clean.

5.1 Desingularizing the logarithmic coefficients

Write $P_i(L) = \sum_{j=0}^{d_i} c_{ij} L^j$. Introduce independent small parameters

$$\zeta_{ik} = u^{\delta_i} L^k, \quad 0 \leq k \leq d_i.$$

In the normalized equation for $v = g/u$, replace the actual combinations $u^{\delta_i} L^k$ by these independent parameters. Expanding $P_i(L + \log v)$ gives the analytic equation

$$v^p \left[1 + \sum_i \sum_{j=0}^{d_i} \sum_{k=0}^j c_{ij} \binom{j}{k} \zeta_{ik} v^{\delta_i} (\log v)^{j-k} \right] = 1. \quad (5.1)$$

Near $v = 1$, fix the analytic logarithm with $\log 1 = 0$; all real powers of v are then analytic. At $\zeta = 0$, the solution is $v = 1$ and the derivative with respect to v is $p \neq 0$.

The analytic implicit-function theorem therefore gives a convergent multivariate Taylor expansion $v = V(\zeta)$ in a fixed polydisc around the origin. Every actual parameter $u^{\delta_i} L^k$ tends to zero as $u \rightarrow 0^+$, so the real curve of parameters eventually lies in that polydisc. The resulting solution is the addressed inverse from Proposition 3.2.

Grouping the absolutely convergent multivariate Taylor series by the number of factors originating from each correction index i gives (4.1). For each fixed multi-index, only finitely many logarithmic choices occur. The coefficient computation in Section 4 identifies the resulting polynomial with (4.2). This proves the convergence assertion in Theorem 4.1, not merely formal consistency.

5.2 A computable remainder for every strict cutoff

For $C > 1$, put $B = C - 1$ and define the finite approximant

$$A_C(y) = u \left(1 + \sum_{0 < \Delta < B} u^\Delta Q_\Delta(L) \right), \quad (5.2)$$

where the summation is over multi-indices and equal powers are collected. Let

$$\delta_* = \min_i \delta_i, \quad d = \max_i \deg P_i, \quad N = \left\lceil \frac{B}{\delta_*} \right\rceil. \quad (5.3)$$

Theorem 5.1 (Uniformizer-power cutoff bound). *For nonempty exact finite input,*

$$g(y) - A_C(y) = O\left(u^C (1 + |\log u|)^{Nd}\right) \quad (y \rightarrow 0^+). \quad (5.4)$$

For an exact monomial, the inverse is exactly u and the remainder is zero.

Proof. Truncate the analytic multivariate Taylor series $V(\zeta)$ at total degree $N - 1$. Its remaining tail is $O(\|\zeta\|^N)$ in a smaller fixed polydisc. Along the actual curve,

$$\|\zeta\| = O\left(u^{\delta_*} (1 + |L|)^d\right),$$

so this tail is $O(u^{N\delta_*} (1 + |L|)^{Nd})$. Since $N\delta_* \geq B$, it is $O(u^B (1 + |L|)^{Nd})$.

Among the finitely many terms of total degree below N , those omitted from (5.2) have weight $\Delta \geq B$ and logarithmic degree at most $(N - 1)d$. Their sum obeys the same bound. No term of total degree N or larger can have weight below B , because $\Delta \geq N\delta_* \geq B$. Multiplying the relative estimate by u proves (5.4). $\square$

The bound is intentionally conservative for mixed generators. It requires no search for the first nonzero omitted shell and survives arbitrary coefficient cancellations. The package returns precisely this bound. In the one-generator logarithmic example with integer cutoff C , it has the sharp logarithmic degree $C - 1$ for the first omitted block.

Corollary 5.2 (Pure-power weakening). *For every fixed $\varepsilon > 0$,*

$$g(y) - A_C(y) = O(u^{C-\varepsilon}).$$

One cannot generally replace this by $O(u^C)$ when logarithmic coefficients occur at the boundary.

5.3 Sharper next-shell estimates

Suppose $\lambda \geq B$ is the least omitted weight with a nonzero *collected* polynomial Q_λ . Local finiteness provides a gap to the next possible weight. Applying Theorem 5.1 at a cutoff inside that gap gives

$$g(y) - A_C(y) = u^{1+\lambda} Q_\lambda(L) + o\left(u^{1+\lambda}(1 + |L|)^{\deg Q_\lambda}\right).$$

If the inverse is already a finite expression and no omitted nonzero block exists, the remainder is instead exactly zero. A program must not claim a nonzero next term merely because the support semigroup allows one.

6 The logarithmic example to arbitrary order

Here $a = p = 1$, $\delta_1 = 1$, and $P_1(L) = L + 1$. Thus

$$g_1(y) = y + \sum_{n \geq 1} y^{n+1} q_n(\log y), \quad q_n(L) = \frac{(-1)^n}{n!} \prod_{j=1}^{n-1} (D_L + n + 1 + j)(L + 1)^n. \quad (6.1)$$

Each q_n has degree n . The first five are

$$\begin{aligned} q_1 &= -L - 1, \\ q_2 &= 2L^2 + 5L + 3, \\ q_3 &= -5L^3 - \frac{41}{2}L^2 - 27L - \frac{23}{2}, \\ q_4 &= 14L^4 + \frac{241}{3}L^3 + \frac{335}{2}L^2 + 151L + \frac{299}{6}, \\ q_5 &= -42L^5 - \frac{3727}{12}L^4 - \frac{5365}{6}L^3 - 1256L^2 - \frac{5177}{6}L - \frac{2789}{12}. \end{aligned} \quad (6.2)$$

Further exact coefficients are included in the machine-readable validation results and can be generated at any chosen cutoff subject to resource limits.

6.1 Catalan numbers in the leading logarithms

The derivative part of $D_L + c$ lowers logarithmic degree, so the leading coefficient of q_n comes from multiplying only the scalar parts:

$$[L^n]q_n(L) = \frac{(-1)^n}{n!} \prod_{j=1}^{n-1} (n + 1 + j) = (-1)^n \frac{1}{n+1} \binom{2n}{n}.$$

The leading logarithms are signed Catalan numbers. Lower logarithmic powers are not optional decorations: their derivatives contribute to higher inverse coefficients. Replacing $\log g$ by $\log y$ too early recovers some leading terms but misses those subleading contributions.

6.2 A compact direct Wolfram construction

The observable homotopy formula immediately gives a minimal coefficient constructor for the identity-leading case:

```
(* h is the correction to the identity, as an expression in z. *)
identityInverseJet[h_, {z_, y_, n_Integer}] :=
  y + Total[Table[
    (-1)^k/k! D[h^k, {z, k - 1}] /. z -> y,
    {k, 1, n}
  ]];

identityInverseJet[z^2 (1 + Log[z]), {z, y, 4}]
```

This finite derivative formula explains why the problem is computationally tractable. It does not by itself implement mixed-weight sorting, input normalization, branch metadata, or a rigorous remainder. The full package adds those layers.

6.3 The remainder can be seen numerically and symbolically

For the two-block approximation $A_3 = y - y^2(1 + \log y)$,

$$\frac{g_1(y) - A_3(y)}{y^3 \log^2 y} \rightarrow 2.$$

The independently computed ratios at $y = 10^{-20}, 10^{-50}, 10^{-100}$ are approximately 1.892841, 1.956797, and 1.978342, respectively. The slow approach reflects the next coefficient $5/\log y$, not a failure of the expansion. The quotient by y^3 itself diverges like $2 \log^2 y$, directly showing why a bare $O(y^3)$ would be incorrect.

7 The irrational-power example and its convergence radius

Consider the entire family

$$f_r(x) = x + x^r, \quad r > 1, \quad \delta = r - 1.$$

Here the logarithmic polynomial is 1, so the differential operators in (4.2) act by their constant parts. For $n \geq 1$,

$$\begin{aligned} c_n &= \frac{(-1)^n}{n!} \prod_{j=1}^{n-1} (1 + n\delta + j) \\ &= \frac{(-1)^n}{n} \binom{nr}{n-1} = \frac{(-1)^n}{1 + n(r-1)} \binom{nr}{n}. \end{aligned} \quad (7.1)$$

With $c_0 = 1$, the inverse is

$$g_r(y) = \sum_{n=0}^{\infty} c_n y^{1+n(r-1)}. \quad (7.2)$$

There is no rationality requirement on r in this identity.

The first coefficients are

$$1, \quad -1, \quad r, \quad -\frac{r(3r-1)}{2}, \quad \frac{r(4r-1)(4r-2)}{6}, \quad \dots$$

For $r = 2$ these reduce to the signed Catalan coefficients of the inverse of $x + x^2$. For $r = \sqrt{2}$, equation (1.8) follows at once.

7.1 A genuine convergent ordinary series in a new parameter

Put $t = y^{r-1}$ and $v = g_r(y)/y$. The equation becomes

$$v + tv^r = 1, \quad (7.3)$$

with $v(0) = 1$. Thus v is an ordinary analytic function of t near zero, even though its expression as a function of y uses irrational powers. The critical point of (7.3) satisfies

$$v + tv^r = 1, \quad 1 + rtv^{r-1} = 0.$$

It occurs at

$$v_* = \frac{r}{r-1}, \quad t_* = -\frac{(r-1)^{r-1}}{r^r}.$$

To establish the radius without assuming that this critical point is the only relevant complex obstruction, use the exact coefficients. Stirling's formula gives

$$|c_n| \sim \frac{\sqrt{r}}{\sqrt{2\pi}(r-1)^{3/2}} n^{-3/2} \left(\frac{r^r}{(r-1)^{r-1}} \right)^n. \quad (7.4)$$

The root test therefore proves the exact radius

$$R_t = \frac{(r-1)^{r-1}}{r^r}. \quad (7.5)$$

The corresponding positive- y threshold for this series representation is $R_y = R_t^{1/(r-1)}$. At $r = \sqrt{2}$,

$$R_t \approx 0.4251956066518408, \quad R_y \approx 0.1268627051233097.$$

The real inverse exists for every $y > 0$, but this particular Taylor series in t does not converge for every such y . Beyond its convergence disk, one may continue the analytic solution or solve the monotone real equation numerically. Global invertibility and the radius of a local representation are different questions.

7.2 A non-asymptotic tail majorant inside the disk

For fixed $r > 1$ and any $0 < T < R_t$, convergence gives a finite constant $M_T = \sum_{n \geq 0} |c_n| T^n$. If $0 \leq t < T$, then

$$\left| v(t) - \sum_{n=0}^{N-1} c_n t^n \right| \leq M_T \left(\frac{t}{T} \right)^N.$$

This bound is not optimized, but it separates a uniform convergence statement from a purely asymptotic order symbol. Sharper alternating-tail bounds require checking the term-magnitude monotonicity at the particular t ; the package does not assume it without a check.

8 Mixed supports and nonunit leading powers

8.1 Irrational powers mixed with logarithms

For

$$f(x) = x + x^{\sqrt{2}} + x^2(1 + \log x),$$

Table 1: The inverse of $x + x^{\sqrt{2}} + x^2(1 + \log x)$ below power 3.

Power of y	Coefficient polynomial in $L = \log y$
1	1
r	-1
$2r - 1$	r
2	$-L - 1$
$3r - 2$	$-3 + r/2$
$1 + r$	$(r + 2)L + r + 3$
$4r - 3$	$-4 + 17r/3$
$2r$	$-(5 + 3r)L - 13/2 - 5r$

write $r = \sqrt{2}$ and $L = \log y$. The relative support is generated by $r - 1$ and 1. The inverse blocks with absolute exponent less than 3 are shown in Table 1. They are already sorted by their exact real exponents.

The first omitted block is $y^3(2L^2 + 5L + 3)$; a sharpened remainder is therefore $O(y^3L^2)$. The universal bound of Theorem 5.1 is more conservative: $O(y^3(1 + |L|)^5)$. Both are valid, but they encode different amounts of support information.

A perturbation-degree cutoff would interleave these terms incorrectly. For example, repeated use of the x^{r-1} correction can produce several terms before a single use of a higher-weight correction. The package therefore enumerates by the inequality $m_1(r - 1) + m_2 < B$, not by a single common bound on $m_1 + m_2$.

8.2 Collisions and exact zero blocks

For $f(x) = x + x^2 + x^3$, relative weights 1 and 2 are dependent. The weight-2 coefficient receives both a repeated weight-1 contribution and a direct weight-2 contribution. One obtains

$$f^{-1}(y) = y - y^2 + y^3 + 0y^4 - 4y^5 + 14y^6 + \dots$$

The vanishing y^4 block is a useful regression test. Keeping one representative multi-index for each exponent would be wrong: the contributions must be summed, not deduplicated before coefficient calculation.

8.3 A nonunit scale changes both the power and the logarithm

Consider

$$f(x) = 4x^2 \left(1 + x \log x + 3x^{\sqrt{2}} \right). \quad (8.1)$$

Put $u = \sqrt{y/4}$ and $L = \log u = \frac{1}{2} \log(y/4)$. Up to absolute u -power 4, the inverse is

$$\begin{aligned} g(y) = & u - \frac{1}{2}Lu^2 - \frac{3}{2}u^{1+\sqrt{2}} + \left(\frac{5}{8}L^2 + \frac{1}{4}L \right) u^3 \\ & + \frac{3}{4}((4 + \sqrt{2})L + 1)u^{2+\sqrt{2}} + \frac{9}{8}(3 + 2\sqrt{2})u^{1+2\sqrt{2}} + O(u^4(1 + |L|)^3). \end{aligned} \quad (8.2)$$

Replacing L by $\log y$ would change coefficients and make this expansion false. The leading coefficient 4 contributes a logarithmic shift as well as a multiplicative scale. This is why the package carries u and L as independent internal symbols until final specialization.

9 Residual checking and the passage to actual inverse errors

9.1 The exact residual identity

Let $A(y) > 0$ be any finite approximation with $A/u \rightarrow 1$. Define

$$R(y) = f(A(y)) - y, \quad \rho(u) = \frac{f(A(y))}{au^p} - 1.$$

These are related by $R = au^p \rho$. The mean-value theorem gives, for a point ξ between A and g ,

$$g(y) - A(y) = -\frac{R(y)}{f'(\xi)}. \quad (9.1)$$

Since $A/u \rightarrow 1$, $g/u \rightarrow 1$, and $f'(x) \sim apx^{p-1}$,

$$g(y) - A(y) = -\frac{u}{p} \rho(u) (1 + o(1)) \quad (9.2)$$

where the ratio is interpreted along nonzero residual values. In particular, if

$$\rho(u) = u^\lambda S(L) + o(u^\lambda (1 + |L|)^{\deg S}), \quad S \neq 0,$$

then the first inverse correction is $-u^{1+\lambda} S(L)/p$. The divisor is p , not 1, unless the leading exponent is one.

This proves that a power-log residual calculation can certify the asymptotic correctness of an approximation when combined with the stated real-branch and derivative hypotheses. A list of formal zero coefficients alone is not a finite-precision interval certificate.

Corollary 9.1 (Global error bounds for the original examples). *For every $y > 0$ and every positive approximation A , one has*

$$\begin{aligned} |g_1(y) - A| &\leq \frac{|A + A^2(1 + \log A) - y|}{1 - 2e^{-5/2}}, \\ |g_2(y) - A| &\leq |A + A^{\sqrt{2}} - y|. \end{aligned}$$

Proof. Apply the mean-value theorem and the global bounds $f'_1 \geq 1 - 2e^{-5/2}$ and $f'_2 \geq 1$ proved in Section 1. No small-argument restriction or root bracket is needed for these two inequalities, beyond $A, y > 0$. $\square$

9.2 The independent sparse-ring checker

The package stores $A/u = 1 + V$ as a sparse set of positive-weight blocks. It independently constructs

$$\rho = (1 + V)^p \left[1 + \sum_i u^{\delta_i} (1 + V)^{\delta_i} P_i(L + \log(1 + V)) \right] - 1. \quad (9.3)$$

Binomial and logarithmic series are evaluated in a clipped sparse ring, using (2.3). Polynomial substitution uses Horner's rule. Each multiplication discards exponents at or beyond the requested relative cutoff. No coefficient formula from (4.2) is reused in this checker.

For an inverse generated with absolute cutoff C , the expected residual vanishes below relative power $C - 1$. The checker reports a Boolean certificate only after the requested residual range covers that whole interval. A shorter range returns an explicit `Missing["InsufficientCutoff"]` rather than an unjustified success. Increasing the residual cutoff can expose the first omitted inverse block via (9.2).

9.3 An explicit finite-point slope bound

There is also a concrete route from a residual to a numerical error bound. For $0 < u < 1/2$, consider $I_u = [u/2, 2u]$ and put $M = |\log u| + \log 2$. Define coefficient majorants

$$\widehat{P}_i(M) = \sum_k |[L^k]P_i|M^k,$$

and, with $T_i = (p + \delta_i)P_i + P'_i$, define $\widehat{T}_i$ similarly. Set

$$\begin{aligned} E_0(u) &= \sum_i (2u)^{\delta_i} \widehat{P}_i(M), \\ E_1(u) &= \sum_i (2u)^{\delta_i} \widehat{T}_i(M). \end{aligned}$$

If

$$E_0(u) < 1 - 2^{-p}, \quad E_1(u) \leq p/2,$$

then $f(u/2) < au^p < f(2u)$ and

$$f'(x) \geq m_u := \frac{ap}{2} u^{p-1} 2^{-|p-1|} \quad \text{for } x \in I_u. \quad (9.4)$$

Indeed, the first inequality bounds $1 + H(x)$ on the bracket; the second bounds the correction to p in (3.2). The minimum of x^{p-1} on this bracket is $u^{p-1} 2^{-|p-1|}$.

Once $A \in I_u$, equation (9.1) gives

$$|g(y) - A(y)| \leq \frac{|f(A(y)) - y|}{m_u}. \quad (9.5)$$

To turn this into a machine-certified numerical bound, all inequalities and function evaluations must themselves use verified enclosures. The package's algebraic residual checker does not silently claim that an ordinary floating-point comparison supplies such an enclosure.

9.4 Newton refinement as a scalable alternative

In the normalized ring, $\partial\mathcal{F}/\partial V$ has constant term p and is a unit. A Newton step

$$V_{\text{new}} = V - \frac{\mathcal{F}(V)}{\partial\mathcal{F}(V)/\partial V}$$

therefore makes sense with clipped series arithmetic. If the error has positive valuation λ , the next error has valuation at least 2λ , because the normalized equation is analytic in V and its linearization is invertible. This gives a Newton–Hensel strategy with increasing cutoffs.

The shipped generator uses the explicit multi-index formula; its independent checker already supplies most of the sparse operations needed for a future Newton backend. Newton generation is described here as an alternative algorithm, not claimed as an additional implemented public API.

10 Input jets, observables, and information below all powers

10.1 When the finite input is an approximation

Suppose F is a prepared finite germ and an actual function satisfies

$$f(x) - F(x) = O\left(x^{p+\beta}(1 + |\log x|)^k\right), \quad \beta > 0, \quad (10.1)$$

with both functions having the addressed inverse branches and $f'(x) \sim apx^{p-1}$. Evaluating f at $F^{-1}(y)$ and using the residual transfer theorem yields

$$f^{-1}(y) - F^{-1}(y) = O\left(u^{1+\beta}(1 + |\log u|)^k\right). \quad (10.2)$$

Thus inverse terms above that uncertainty scale are not determined by the supplied input jet alone. The package interprets its finite expression as exact; a user applying it to a truncated jet must combine its output remainder with (10.2).

A value asymptotic does not automatically imply a derivative asymptotic. For example,

$$f(x) = x + x^4 \sin(x^{-4}) = x + O(x^4)$$

has

$$f'(x) = 1 + 4x^3 \sin(x^{-4}) - 4x^{-1} \cos(x^{-4}),$$

which is not eventually positive. One cannot infer an inverse branch merely by differentiating the written order symbol. Exact finite prepared input avoids this problem because its derivatives are explicit.

10.2 Powers and logarithms of the inverse

Replace the observable $\Phi(Z) = Z^{1/p}$ in (4.3) by $Z^{q/p}$. For fixed real q , the same calculation gives

$$g(y)^q = u^q \left(1 + \sum_{m \neq 0} u^\Delta \frac{q(-1)^n}{p^n m!} \prod_{j=1}^{n-1} (D_L + q + \Delta + pj) P^m(L) \right). \quad (10.3)$$

For $q = 0$ the correction vanishes and the identity is $1 = 1$. Differentiating this identity with respect to q at zero gives

$$\log g(y) = L + \sum_{m \neq 0} u^\Delta \frac{(-1)^n}{p^n m!} \prod_{j=1}^{n-1} (D_L + \Delta + pj) P^m(L). \quad (10.4)$$

These are useful when the quantity of interest is an observable of the inverse rather than the inverse itself. They also provide independent symbolic consistency tests.

For a stored inverse block $u^e Q(L)$,

$$\frac{d}{dy}(u^e Q(L)) = \frac{1}{ap} u^{e-p} (e + D_L) Q(L).$$

Termwise differentiation is justified within the analytic finite-input construction on an appropriate local sector; it is not licensed merely by a bare real-axis big- O remainder for an arbitrary function.

10.3 Flat perturbations have separate inverse effects

Let $\tilde{f} = f + \varepsilon$ where ε is smaller than every power of x , and suppose the perturbed function retains the addressed branch and the required derivative control. Then

$$\tilde{g}(y) - g(y) = -\frac{\varepsilon(g(y))}{f'(g(y))} (1 + o(1))$$

under the corresponding stability hypotheses, for example when the perturbation and its derivative are uniformly negligible on the small comparison interval. Such an effect is invisible to every finite power-log cutoff. This is not a failure of reversion; it is information not present in the chosen formal scale.

For instance, x and $x + e^{-1/x}$ have the same pure-power asymptotic expansion at zero, but their inverses differ by $-e^{-1/y}(1 + o(1))$. The perturbation is monotone and its derivative is flat, so the real-germ comparison is well behaved. An exponential sector would be needed to record this difference explicitly.

11 Leading logarithmic pivots and other endpoints

11.1 The exact inverse of a leading log-power monomial

The core class excludes a nonconstant logarithmic *leading* block. That case should be normalized differently, not forced into a correction of positive power valuation. Consider

$$f_0(x) = ax^p(-\log x)^q, \quad 0 < x < 1, \quad a, p > 0. \quad (11.1)$$

With $t = -\log x$ and $Y = y/a$, the equation is

$$Y = t^q e^{-pt}.$$

For $q \neq 0$, this gives

$$b(y) = \exp \left[\frac{q}{p} W_k \left(-\frac{p}{q} \left(\frac{y}{a} \right)^{1/q} \right) \right]. \quad (11.2)$$

The branch is forced by $t \rightarrow +\infty$:

$$k = -1 \quad (q > 0), \quad k = 0 \quad (q < 0).$$

For $q = 0$, use $b(y) = (y/a)^{1/p}$. The real branches of Lambert W and their domains are described in DLMF §4.13 [4].

For $q > 0$, the small- x branch has $0 < Y < (q/(pe))^q$. The principal branch W_0 would instead give $t \rightarrow 0$ and $x \rightarrow 1$, another genuine real inverse germ of the same leading model. For $q < 0$, the argument of W_0 is positive and tends to infinity, yielding the desired $t \rightarrow \infty$ branch. This branch distinction is precisely why `LogPowerBaseInverse` is a separate helper.

11.2 The logarithm-of-logarithm scale

Put $H = \log(a/y)$. The equation $pt = H + q \log t$ gives, by successive substitution,

$$t = \frac{1}{p} \left[H + q \log \frac{H}{p} + \frac{q^2}{H} \log \frac{H}{p} + O\left(\frac{(\log H)^2}{H^2}\right) \right]. \quad (11.3)$$

Consequently,

$$b(y) = \left(\frac{y}{a} \right)^{1/p} \left(\frac{H}{p} \right)^{-q/p} \left[1 - \frac{q^2}{pH} \log \frac{H}{p} + O\left(\frac{(\log H)^2}{H^2}\right) \right]. \quad (11.4)$$

The relative small quantities are now inverse logarithms with logarithm-of-logarithm coefficients. These are not positive powers of the uniformizer in Definition 3.1. The formula identifies the necessary extension of scale instead of pretending that the existing finite polynomial-log representation already covers it.

11.3 Perturbing an exactly invertible pivot

Suppose $f = f_0 + h$ and the addressed inverse $b = f_0^{-1}$ is known. In the coordinate $s = f_0(x)$ the equation becomes

$$y = s + \eta(s), \quad \eta(s) = h(b(s)), \quad x = b(s).$$

The observable formula gives the homotopy jet

$$x = b(y) + \sum_{n=1}^N \frac{(-1)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} (b'(y)\eta(y)^n). \quad (11.5)$$

This is implemented by `InversePerturbationJet`. The argument supplied as the perturbation is η in the *pivot coordinate*, not the original $h(x)$. Establishing the asymptotic ordering, convergence, or remainder of such a jet requires the relevant prepared-scale hypotheses; the helper deliberately does not attach an automatic big- O claim.

11.4 Shifts, the other real side, and infinity

For a finite endpoint x_0 with $f(x_0) = y_0$, first introduce $x = x_0 + t$ and $y = y_0 + s$ and address the intended side $t \rightarrow 0^+$ or $t \rightarrow 0^-$. Real noninteger powers on a negative side require an explicit real-valued definition; a complex principal power must not be silently reinterpreted as a real one.

For an inverse at infinity, a useful transformation is

$$F(t) = \frac{1}{f(1/t)}.$$

If $G = F^{-1}$ is the addressed inverse near zero, then

$$f^{-1}(y) = \frac{1}{G(1/y)}.$$

The transformed function may have an infinite or rational power-log expansion rather than exact finite input. In that case one must prepare a finite jet and propagate its uncertainty as in Section 10. These endpoint transformations are mathematical recipes, not automatic preprocessing implemented by the core parser.

12 Wolfram Language package: interface and implementation

12.1 Loading and solving the original examples

After extracting the archive, load the package from its actual path:

```
Get["/path/to/PowerLogInverse/Kernel/PowerLogInverse.wl"];

PowerLogInverse[
  x + x^2 (1 + Log[x]), {x, y, 5}
]

PowerLogInverse[
  x + x^Sqrt[2], {x, y, 4 Sqrt[2] - 3}
]
```

These requests mean strict absolute uniformizer-power cutoffs. Since $a = p = 1$ in both examples, the uniformizer is just y . The second call retains exactly the four blocks in (1.8).

For the coefficients and error metadata together:

```
d = PowerLogInverseData[
  x + x^2 (1 + Log[x]), {x, y, 5}
];
d["Expression"]
d["Remainder"]
(* PowerLogOrder[y, 5, 4] *)

PowerLogResidual[d]["Certificate"]
(* Expected in a native kernel: True *)

PowerLogResidual[d, 5]["Expression"]
(* First nonzero relative residual blocks, below u-power 5. *)
```

The finite expression is the approximation only. It is not an expression with an order object added to it. The descriptor `PowerLogOrder[u, C, K]` means $O(u^C (1 + |\log u|)^K)$ as $u \rightarrow 0^+$. No algebraic rules for adding, multiplying, or composing such descriptors are installed.

12.2 Public functions

Function	Contract
<code>PowerLogInverse</code>	Parse and normalize a finite input expression; return its finite positive-real inverse approximation.
<code>PowerLogInverseData</code>	Return the same approximation with exact support blocks, normalization rules, remainder, input conditions, and resource metadata.
<code>PowerLogReversion</code>	Low-level prepared-input interface $[a, p, \{\{\delta_i, P_i(L)\}\}, \{u, L\}, C]$; returns structured data.
<code>PowerLogResidual</code>	Independently expand the normalized residual of the stored finite approximant through a relative cutoff.
<code>PowerLogOrder</code>	Inert meaning-bearing remainder descriptor, not a series arithmetic head.
<code>LogPowerBaseInverse</code>	Return the exact leading-log pivot inverse with the selected real <code>ProductLog</code> branch.
<code>InversePerturbationJet</code>	Return a finite observable Lagrange homotopy jet for a supplied pivot coordinate; no automatic remainder.

The low-level interface is useful when normalization is already known:

```
d = PowerLogReversion[
  4, 2, {{1, L}, {Sqrt[2], 3}}, {u, L}, 4
];
d["Terms"]
PowerLogResidual[d]["Certificate"]
```

Here `d["Expression"]` still contains the independent symbols u, L . Specialize them to $u = \sqrt{y}/4$ and $L = \frac{1}{2} \log(y/4)$ only when an expression in y is desired. The high-level interface performs that specialization and retains `UniformizerRule` and `LogRule`.

12.3 The parser's deliberate domain

The input must expand into a finite sum of x^r times polynomials in $\text{Log}[x]$. Exponents, including the leading p and the cutoff, must be exact real algebraic numbers. Coefficients must be exact and provably real constants. The smallest power must have a constant positive coefficient, and its exponent must be positive.

The parser rejects a nonconstant leading logarithmic block, inverse logarithmic powers, unprepared functions such as $\sin x$, unresolved symbolic exponents, and machine-precision exponents. These restrictions are explicit interface boundaries, not claims that the corresponding mathematical problems lack asymptotic expansions. A symbolic parameter can be specialized before calling the package, or one can use the general homotopy helper when its hypotheses have been established separately.

A `ConditionalExpression` is accepted only when its condition is provably implied by $x > 0$. An unresolved condition is returned as a `Failure`; it is not discarded. Eventual conditions such as $x < 1$ that are not globally implied by $x > 0$ require explicit germ preparation. Use distinct source and target symbols, with the target absent from the input expression.

No use is made of `PowerExpand`. Identities involving logarithms and powers are derived from positivity and the stated normalization, not from unconditional manipulation of principal complex branches. Likewise, no irrational exponent is replaced by a rational approximation. If r is changed to $r + \epsilon$, the ratio $x^{r+\epsilon}/x^r = \exp(\epsilon \log x)$ need not remain close to 1 in the asymptotic limit, no matter how small a fixed nonzero ϵ is.

12.4 Exact exponent keys and collisions

Exponent arithmetic is reduced with `RootReduce`. Rows are sorted by exact inequalities and adjacent equal exponents are combined using exact equality tests. This makes collision handling independent of accidental syntactic differences between equivalent algebraic numbers. An undecided order is a failure, not an invitation to guess from decimal approximations. Logarithmic polynomial coefficients are expanded and simplified before zero blocks are removed.

Internally, a row is `{exponent, polynomial}`. Lists rather than symbolically keyed associations are used for the sparse arithmetic. Associations provide the outer API metadata. The residual checker uses the stored `Terms` and prepared corrections; it does not infer new coefficients from the already computed inverse formula.

12.5 Enumeration and resource controls

For the requested relative cutoff B , the generator enumerates only the integer simplex $m \cdot \delta < B$. Generators whose weight is already at least B can be omitted from coefficient enumeration, though they remain part of the input model and its remainder analysis. The implementation uses an iterative depth-first traversal, avoiding a dependence on the Wolfram kernel's recursion-depth setting.

The public option names are strings. Their defaults are

Option	Default
"MaxMultiIndices"	100000
"MaxTerms"	50000
"MaxTotalDegree"	256
"MaxProducts"	2000000

The product limit also bounds a growing enumeration frontier. Exceeding a budget returns a named `Failure`, not a partially computed expression masquerading as a complete truncation. These limits do

not constitute wall-clock guarantees: exact algebraic simplification and coefficient expression growth can still dominate a calculation.

For fixed s and positive generators, the number of visited terminal indices is bounded by a finite box and grows like the volume of a simplex, $B^s / (s! \prod_i \delta_i)$, at large cutoffs. When generators are dependent, the number of distinct output exponents can be much smaller than the number of contributing multi-indices. Direct coefficient enumeration is transparent and useful at moderate cutoffs; a sparse Newton backend is an appropriate optimization for large, highly colliding problems.

12.6 Implementation map

The core file separates the mathematical and operational layers:

1. Exact-domain validation, polynomial parsing, positive leading-term selection, and condition checking.
2. Cutoff-finite multi-index enumeration and the differential-polynomial coefficient formula.
3. Exact exponent collection, finite-expression construction, and the analytic remainder metadata from Theorem 5.1.
4. Independent clipped sparse addition, multiplication, binomial powers, logarithms, polynomial substitution, and residual checking.

The last two public helpers are intentionally outside the finite core: they expose leading-log pivots and observable homotopy derivatives without pretending that their general remainder problem has already been solved by the core parser.

13 Verification evidence and reproducibility

13.1 What was executed

The archive contains an independent Python implementation using exact SymPy algebra and high-precision mpmath arithmetic. It passed **46 of 46 checks**. The checks cover the explicit logarithmic coefficients, generalized Fuss–Catalan coefficients, ordinary Catalan reversion, strict irrational cutoff boundaries, mixed supports, dependent and duplicate generators, nonunit and fractional leading exponents, exact residual cancellation, deliberate coefficient corruption, and numerical improvement under successive truncation.

The Python coefficient generator uses a bounded Cartesian enumeration, whereas the Wolfram implementation uses a simplex depth-first traversal. The Python residual checker uses clipped binomial and logarithmic composition rather than the coefficient formula. An additional ordinary integer-power substitution checks the logarithmic example independently of that sparse residual code. These checks test the mathematics by several routes; they do not amount to execution of the Wolfram source.

The archive also contains **40 native Wolfram Language regression test definitions**. Native Wolfram-kernel execution was not available in the build environment. The Wolfram files passed a lexical audit for balanced delimiters, strings, and nested comments, but that audit is not a Wolfram parser or evaluator. Accordingly, native execution is recorded as *unverified*, not as a successful test run.

13.2 Numerical results

Table 3 reports absolute inverse errors at $y = 10^{-8}$. Roots were obtained independently by high-precision bisection in a real bracket. These are numerical diagnostics, not interval-certified error bounds. The

exact coefficients were converted to the requested precision only at evaluation time.

Table 3: Independently measured errors at $y = 10^{-8}$. Cutoffs refer to u .

Input function	Strict cutoff C	Absolute inverse error
$x + x^2(1 + \log x)$	3	$5.8953980 \times 10^{-22}$
	5	$1.1639260 \times 10^{-34}$
	7	$3.0778899 \times 10^{-47}$
$x + x^{\sqrt{2}}$	$1 + 2(\sqrt{2} - 1)$	$3.3323930 \times 10^{-15}$
	$1 + 4(\sqrt{2} - 1)$	$2.2301916 \times 10^{-21}$
	$1 + 6(\sqrt{2} - 1)$	$1.8441485 \times 10^{-27}$
Equation (8.1)	3	$7.2770394 \times 10^{-12}$
	4	$5.4503639 \times 10^{-15}$
	5	$4.5863230 \times 10^{-18}$

The first two-block irrational approximation is relatively less accurate than the two-block logarithmic approximation at the same y . This is expected: its first relative small scale is $y^{\sqrt{2}-1}$, which decreases much more slowly than $y|\log y|$. Counting blocks alone is not a fair measure of asymptotic accuracy across different supports.

13.3 Reproducing the computations

From the archive root, the independent validation commands are

```
python Validation/validate.py
python Validation/static_check.py
```

The former regenerates `validation-results.json`, including numerical errors and coefficients; the latter regenerates `static-results.json`. The Python requirements are SymPy and mpmath. Versions used in the executed run are recorded in the JSON file.

For a native Wolfram installation:

```
wolframscript -file Tests/run-tests.wls
wolframscript -file Examples/examples.wls
```

Alternatively, evaluate `TestReport` on the supplied `.wlt` file in a notebook. The test suite includes failure-path tests in addition to coefficient comparisons. In particular, a corrupted inverse block must make the independent residual certificate false, and an insufficient residual range must not return true.

Compile this self-contained article source with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  real_branch_reversion.tex
```

The archive includes the compiled PDF, so no TeX installation is needed to read the article.

14 Scope and conclusions

The two motivating inverses are covered to arbitrary finite order by the same coefficient calculus. Their apparent incompatibility with ordinary series reversion comes from representation and branch selection,

not from a lack of constructive formulas. The first needs logarithmic polynomials at integer powers; the second needs irrational powers; their mixtures need a finitely generated positive support semigroup.

The finite-input result is stronger than a formal expansion: sufficiently near the selected positive endpoint, the constructed power-log series actually converges. Its strict cutoff has a proved, computable logarithmic remainder envelope. An independently computed normalized residual provides another route to asymptotic verification and, with a verified slope bound, to finite-point error control.

The current executable core is deliberately exact and finite. It does not automatically discover arbitrary transseries scales, decide symbolic parameter orderings, invert every leading logarithmic polynomial, classify all global branches, or certify floating-point intervals. The article identifies concrete extensions rather than hiding those tasks inside an unqualified universal inverse command. Leading logarithmic monomials have an implemented real- W pivot helper; general pivot perturbations have an implemented homotopy-jet helper; the broader scale and residual hypotheses remain explicit.

The essential computational contract is simple: specify the real germ, prepare the leading term, enumerate the actual ordered support, collect all equal-weight contributions, and attach only a remainder justified by that preparation. This is sufficient to resolve both original examples and a substantial mixed power-logarithmic class without invoking an unspecified inverse branch.

A A coefficient-level pseudocode specification

The following pseudocode describes the generator independently of Wolfram Language evaluation details. It is also a compact reference for a port to another exact symbolic system.

```

Input:  $a > 0$ ,  $p > 0$ , corrections  $(\delta_i, P_i)$ , cutoff  $C > 1$ 
Normalize duplicate deltas; remove zero  $P_i$ 
 $u := \text{positive}(y/a)^{1/p}$ ;  $L := \text{Log}(y/a)/p$ 
 $B := C - 1$ 
 $\text{rows} := [(1, 1)]$ 

for each nonzero multi-index  $m$  with  $\text{dot}(m, \delta) < B$ :
   $n := \text{sum}(m)$ 
   $\Delta := \text{dot}(m, \delta)$ 
   $Q := \text{product}(P_i(L)^{m_i})$ 
  for  $j := 1$  to  $n-1$ :
     $Q := \text{derivative}(Q, L) + (1 + \Delta + p*j)*Q$ 
   $Q := (-1)^n * Q / (p^n * \text{product}(\text{factorial}(m_i)))$ 
  append  $(1 + \Delta, Q)$  to rows

Sort rows by exact real exponent
Add every polynomial sharing the same exponent
Remove exact zero blocks
Return  $\text{sum}(u^e * Q(L) \text{ for } (e, Q) \text{ in rows})$ 

If corrections are nonempty:
   $N := \text{ceiling}(B / \min(\delta_i))$ 
   $K := N * \max(\text{degree}(P_i))$ 
   $\text{remainder} := O(u^C * (1 + \text{abs}(\text{Log}(u)))^K)$ 
Else:
   $\text{remainder} := 0$ 

```

An implementation must test both domain predicates and cutoff comparisons before relying on this specification. Algebraic real exponents supply an exact decision domain. Numerical approximations of these comparisons change the specification and are not used by the shipped package.

B The first logarithmic coefficient recurrences as a check

For the identity-leading logarithmic example, write $g = y + \sum_{n \geq 1} q_n(L)y^{n+1}$. Let $V = g/y - 1$. Then

$$0 = V + y(1 + V)^2(1 + L + \log(1 + V)).$$

The coefficient of y gives $q_1 = -(1 + L)$. At the next relative power,

$$q_2 + (2L + 3)q_1 = 0,$$

so $q_2 = (2L + 3)(L + 1) = 2L^2 + 5L + 3$. One order further,

$$q_3 + (2L + 3)q_2 + \left(L + \frac{5}{2}\right)q_1^2 = 0,$$

which gives the q_3 displayed in (6.2). These equations come directly from substitution and provide a transparent low-order check on the differentiated Lagrange formula.

C Archive contents and provenance

Path	Purpose
article/real_branch_reversion.tex	Self-contained article source.
article/real_branch_reversion.pdf	Compiled article.
Kernel/PowerLogInverse.wl	Main Wolfram Language implementation.
Kernel/init.m	Standard package loader.
Tests/PowerLogInverse.wlt	Native Wolfram regression definitions.
Tests/run-tests.wls	Native test runner with an exit status.
Examples/examples.wls	Worked package calls and a numerical diagnostic.
Validation/validate.py	Executed independent exact/numerical checks.
Validation/validation-results.json	Reproducible results, coefficients, and versions.
Validation/static_check.py	Lexical delimiter audit, not a native parser.
Validation/static-results.json	Audit results.
Sources/question.md	Text of the user-supplied archived question.
README.md, VALIDATION.md	Usage and verification-status notes.
LICENSE	MIT No Attribution license for the newly prepared material.

The mathematical source of the problem is the user-supplied archive of Mathematica Stack Exchange question 236367. The ProveIt corpus was consulted for its formal/analytic architecture, particularly the inversion volume's attention to support, branches, and residual estimates. Its main source file was identified by Git blob aec3c70d4793e286b462e7d8894dede7c754995e during this work. No claim is made that the entire evolving corpus was audited or that the package is a formal proof-assistant implementation of it.

References

- [1] Vladimir Reshetnikov, *How to get an asymptotic of the real-valued branch of the inverse function?* Mathematica Stack Exchange, question 236367. User-supplied archive consulted on 7 September 2026. <https://mathematica.stackexchange.com/q/236367>.
- [2] Vladimir Reshetnikov and contributors, *ProveIt: series-and-transseries draft corpus*, particularly `Transseries_And_Inversion/transseries_and_inversion.tex`. Working repository, consulted 7 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.
- [3] NIST Digital Library of Mathematical Functions, §1.10(vii), *Inverse Functions*, including the Lagrange inversion and extended inversion theorems. <https://dlmf.nist.gov/1.10#vii>.
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- [6] Wolfram Research, *SeriesData* and *InverseSeries*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/SeriesData.html>; <https://reference.wolfram.com/language/ref/InverseSeries.html>.
- [7] Wolfram Research, *InverseFunction*, Wolfram Language documentation, consulted 7 September 2026. <https://reference.wolfram.com/language/ref/InverseFunction.html>.